

Project Proposal Preview

Request: Mere website ke liye responsive search bar banao

1. Requirements

Implement a responsive search bar for the website.

Functional requirements

- Search bar appears in header and collapses into icon on mobile.
- Input field accepts text and triggers search on Enter or icon click.
- Autocomplete suggestions appear as user types, fetched from /search/suggestions API.
- Search results page displays results with pagination.
- Search bar supports keyboard navigation (arrow keys, enter).
- Search bar retains last query in local storage for quick access.

Non-functional requirements

- Responsive across mobile, tablet, desktop.
- Load time < 200ms for suggestions.
- Accessible (WCAG 2.1 AA) with ARIA labels.
- Compatible with latest Chrome, Firefox, Safari, Edge.
- Does not block page rendering (async loading).
- Uses CSS flexbox/grid for layout.

Edge cases

- Empty input triggers no results and shows placeholder.
- API error returns graceful fallback message.
- Very long query string handled without breaking layout.
- Network latency > 2s still shows loading spinner.
- User types quickly; debounce to limit API calls.
- Search bar hidden when user scrolls down (optional).

Assumptions

- Backend API exists at /search/suggestions and /search/results.
- API returns JSON with suggestions array.
- Site uses standard HTML/CSS/JS (no framework specified).
- User has internet connectivity.
- Search bar will be placed in header.
- No authentication required for search.

2. Architecture

Vanilla JS modular implementation with async fetch, debounce and ARIA compliant markup, styled via flexbox/grid for full responsiveness

Component	Responsibility	Type
SearchBar	Render header bar, handle input events, toggle icon on mobile, manage UI state	frontend

Component	Responsibility	Type
SuggestionService	Debounce fetch of /search/suggestions, return suggestion list	frontend
StorageService	Persist last query in localStorage and restore on load	frontend
ResultsPage	Parse query param, fetch /search/results, render paginated list	frontend

Data flow

- HTML loads and asynchronously loads search.js
- SearchBar renders in header, collapses to icon on small viewports
- User types → SuggestionService debounces → fetch suggestions → dropdown updates
- User selects suggestion or presses Enter → query stored via StorageService → browser navigates to /search?query=...
- ResultsPage reads query param → fetches /search/results → renders results with pagination

File structure

- index.html
- assets/css/search.css
- assets/js/search.js
- assets/js/suggestionService.js
- assets/js/storageService.js
- search.html

3. Tools

Tool	Category	Purpose	Required
normalize.css	frontend	CSS reset for consistent styling across browsers	Yes
lodash.debounce	frontend	Debounce function to limit API calls during typing	Yes
Heroicons	frontend	Free SVG icon set for search icon and UI elements	Yes

4. Cost Estimate (tools/services only)

Tool	Pricing model	Est. cost	Notes
normalize.css	free	\$0	free, no usage limits
lodash.debounce	free	\$0	free, no usage limits
Heroicons	free	\$0	free, no usage limits

Estimated monthly total: \$0

Note: These are third-party tool/service cost estimates only, not a cost of building the feature